

Q1. Peak Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = [1, 3, 20, 4, 1, 0]
for i in range(len(numbers)):
    if (i == 0 or numbers[i] >= numbers[i - 1]) and (i == len(numbers) - 1 or numbers[i] >= numbers[i + 1]):
        print(numbers[i])
        break
```

OUTPUT

Q2. Second Largest and Second Smallest - Tuple

Score: 0.00 TC: 0/1 passed

CODE

```
numbers = (10, 20, 5, 40, 15, 30)
sorted_tuple = tuple(sorted((numbers)))
second_smallest = sorted_tuple[1]
second_largest = sorted_tuple[-2]
print(second_smallest)
print(second_largest)
```

INPUT 10 20 5 40 15 30

OUTPUT 10
30

Q3. Frequency of Each Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = [1, 2, 3, 2, 4, 1, 5, 2]
freq = {}
for x in numbers:
    freq[x] = freq.get(x, 0) + 1
for key, value in freq.items():
    print(f"{key} = {value}")
```

OUTPUT

Q4. Missing Number - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = (1, 2, 3, 5, 6)
n = max(numbers)
missing = set(range(1, n + 1)) - set(numbers)
print(missing.pop())
```

OUTPUT

Q5. Majority Element -List

Score: 10.00 TC: 1/1 passed

CODE

```
def find_majority_element(numbers):
    n = len(numbers)
    for num in set(numbers):
        if numbers.count(num) > n // 2:
            return num
    return None

numbers = list(map(int, input().split()))
result = find_majority_element(numbers)

if result is not None:
    print(result)
else:
    print("No majority element")
```

INPUT

2 2 1 2 3 2 2

OUTPUT

Q6. Sum & Average - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = tuple(map(int, input().split()))
total = sum(numbers)
average = total / len(numbers)
print(total)
print(average)
```

INPUT

10 20 30 40 50

OUTPUT

150
30.0

Q7. Element Exists -Tuple

Score: 10.00 TC: 2/2 passed

CODE

```
numbers = tuple(map(int, input().split()))
element = int(input())
if element in numbers:
    print("Element found")
else:
    print("Element not found")
```

INPUT

10 20 30 40 50
30

OUTPUT

Q8. Maximum Element - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = (12, 45, 8, 67, 23)
largest = max(numbers)
print(largest)
```

OUTPUT

7/31/26, 2:04 PM

Student Submissions — Class Work (30.7.26)

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Q9. Lucky Number

Score: 10.00 TC: 2/2 passed

CODE

```
def sum_of_digits(n):  
    if n == 0:  
        return 0  
    return (n % 10) + sum_of_digits(n // 10)  
def is_lucky(n):  
    if n < 10:  
        return n == 7  
    return is_lucky(sum_of_digits(n))  
num = int(input().strip())  
if is_lucky(num):  
    print("Lucky Number")  
else:  
    print("Not a Lucky Number")
```

INPUT

OUTPUT

Q10. Remove Duplicate Elements - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
numbers = (10, 20, 30, 20, 40, 10, 50)  
unique_tuple = tuple({x for x in numbers})  
print(unique_tuple)
```

OUTPUT